UNIT 2 · PROBABILITY · TWO-DAY LAB

Casino Lab — Why the House Always Wins

AP Statistics 2026–27 · Day 1 Tue Nov 10 · Day 2 Thu Nov 12 · analysis due Mon Nov 16

NAME _____ PARTNER _____ TABLE # _____

The big question. Every game in this lab is rigged — slightly. The house edge is only 3–6 cents per chip. So how many rounds does it take before you can actually *see* the edge? You will answer it three ways: **play** a small sample by hand, **simulate** thousands of rounds in Google Sheets, and **calculate** the exact probabilities. Along the way you will find out why dice are independent, why cards are not, and what happens when a simulation gets that wrong.

The two days

DAY 1 · TUE NOV 10 · 80 MIN

Play, then simulate dice

Warm-up: independence refresher (page 2).

Play all three games at your table with your partner: 20 dice rounds, 12 blackjack hands, 12 poker hands. Record every round.

Pool: add your table's totals to the class board.

Build the 10,000-roll dice simulation in Sheets.

DAY 2 · THU NOV 12 · 80 MIN

Simulate cards, then compare

Poker: score 5,000 hands, find the expected value, and catch the "bad deal."

Blackjack: compare two strategies on the same 5,000 hands.

Dashboard: table vs. class vs. simulation vs. theory.

Start the analysis questions; finish them for homework.

House rules (all three games)

- **One partner is the Player, the other is the House.** Swap roles halfway through each game.
- **Every round, the Player bets exactly 1 chip.** No raising, folding, or choices — every rule below is mechanical, so every round is a chance process we can model.
- **Record the Player's net result** for every round: +2, +1, 0 (push — the chip comes back), or −1. Chips just make it feel real; **your record is the data.**
- Each Player starts with 20 chips. If the House runs short, write an IOU and keep playing.

Warm-up — independence refresher

Conditional probability. `P(A | B)` is the probability of A *given that* B happened:

```
P(A | B) = P(A and B) / P(B)
```

In a table or simulation: count the trials where B happened, then find the fraction of *those* where A also happened.

Independent events. A and B are independent if knowing B happened does not change the chance of A:

```
P(A | B) = P(A)
```

Then (and only then) `P(A and B) = P(A)·P(B)`. Dice have no memory. A deck *does* — every card dealt is a card that is gone.

W1 Two dice

INDEPENDENCE

Roll two fair dice. Find `P(die 2 = 6)` and `P(die 2 = 6 | die 1 = 6)`. Are “die 1 is a 6” and “die 2 is a 6” independent? Justify with the definition.

W2 Two cards

WITHOUT REPLACEMENT

Deal two cards from a shuffled 52-card deck (4 aces). Find `P(card 2 is an ace | card 1 is an ace)` and `P(card 2 is an ace | card 1 is not an ace)`. Are the events independent? What is different from W1?

W3 A shortcut that breaks

SIMULATION DESIGN

A student simulates one poker hand by typing `=RANDBETWEEN(1,52)` five times, where 1–52 stand for the 52 cards. Describe a hand this method can produce that is impossible with a real deck. Predict: will pairs come up *more* or *less* often than in a real deal? Why?

W4 Gut check

PREDICTION

Skim the three games on the next pages. Before playing, predict which one gives the House the *biggest* edge, and why.

Game 1 — Dice: The Field

The Player rolls two dice and the **sum** decides everything. This is a real casino bet (the craps "field").

SUM OF THE TWO DICE	PLAYER'S NET
2 or 12	+2
3, 4, 9, 10, or 11	+1
5, 6, 7, or 8	−1

> **Looks generous?** The Player wins on seven different sums and the House on only four. Hold that thought — count the *ways* each sum can happen before you decide who is favored.

Play 20 rounds; swap Player and House after round 10. Write the sum on top, the net below.

Round	1	2	3	4	5	6	7	8	9	10
Sum										
Net										
Round	11	12	13	14	15	16	17	18	19	20
Sum										
Net										

Rounds _____ Total net _____ Average net per round _____

Game 2 — Blackjack: Mimic the Dealer

Card values: 2–10 count face value; J, Q, K count 10; an Ace counts 11 unless that would put the hand over 21, then it counts 1.

1. Shuffle the full deck before *every* hand.
2. Deal two cards to the Player, two to the House (one face up).
3. **Player:** takes cards until the total is **17 or more**. No choices.
4. Player over 21 = **bust: −1 immediately**, even if the House would bust later.
5. **House:** takes cards until its total is **17 or more** (it stands on every 17).

RESULT	PLAYER'S NET
Player busts	−1
Player has a **natural** (Ace + a 10-value card as the first two cards) and the House does not	+2
House busts, or Player's total is higher	+1
Same total (including natural vs. natural)	0
House's total is higher	−1

Both sides follow the *same* rule. Keep an eye out for where the House's edge comes from.

Hand	1	2	3	4	5	6	7	8	9	10	11	12
Player total												
House total												
Net												

Hands _____ Total net _____ Average net per hand _____

Game 3 — Five-Card Deal (poker against a paytable)

Classic two-player poker depends on betting and bluffing, so it has no single probability model. This version is pure chance: **shuffle, deal the Player five cards face up, no draws, no decisions.** The hand is paid from the table. Swap roles after 6 hands.

HAND	WHAT IT MEANS
One pair	two cards of the same rank
Two pair	two different pairs
Three of a kind	three cards of the same rank
Straight	five ranks in a row, any suits (A can be low A-2-3-4-5 or high 10-J-Q-K-A)
Flush	all five cards the same suit
Full house	three of a kind plus a pair
Four of a kind	four cards of the same rank
Straight flush	a straight that is also a flush

HAND (BEST ONE COUNTS)	PLAYER'S NET
No pair (nothing above)	−1
One pair	0 (push)
Two pair	+4
Three of a kind	+8
Straight	+10
Flush	+12
Full house	+15
Four of a kind	+25
Straight flush	+50

Big payouts, rare hands. Which hand do you think matters most to the Player's long-run average?

Hand	1	2	3	4	5	6	7	8	9	10	11	12
Hand name												
Net												

Hands _____ Total net _____ Average net per hand _____

Pool your data. When all three games are done, add your table's *rounds* and *total net* for each game to the class board. Copy the class totals here — you'll need them on the Dashboard.

CLASS TOTALS	DICE	BLACKJACK	POKER
Rounds			
Total net			
Average net per round			

Day 1 build — 10,000 dice rolls

Get the starter. Download `casino-lab-starter.xlsx` from the course page → in Google Drive choose **New › File upload** → open it with Google Sheets → **File › Save as Google Sheets**. Share it with your partner. Color key: `yellow` you build · `gray` pre-built engine, don't edit · `blue` settings you may change.

Heads-up: every edit re-rolls every random number in the workbook. That is not a bug — each recalculation is a brand-new simulation. Press Ctrl/Cmd + R in Sheets to re-roll on purpose.

1. **One roll, row 2** on the `Dice` tab:

```
B2  =RANDBETWEEN(1,6)        C2  =RANDBETWEEN(1,6)        D2  =B2+C2
E2  =IF(OR(D2=2,D2=12),2,IF(AND(D2>=5,D2<=8),-1,1))
```

   Read E2 aloud and check it against the payout table. Then the running record:

```
F2  =E2          G2  =F2/A2
```

2. **Row 3:** select `B2:G3` and press Ctrl/Cmd + D to copy row 2 down. Then fix the running total so it *adds on*:

```
F3  =F2+E3
```

3. **Fill to 10,000 rolls:** copy `B3:G3`, click the Name Box (left of the formula bar), type `B4:G10001`, press Enter, paste. Spot-check three random rows by hand.

4. **Distribution table** (columns I–N). In `J4:J14` type how many of the 36 equally likely (die 1, die 2) outcomes give each sum. Then

```
K4  =J4/36        M4  =COUNTIF($D$2:$D$10001,I4)        N4  =M4/10000
```

   and fill K, M, N down to row 14. Row 15 totals should read 36, 1, 10000, 1.

5. **Expected value vs. simulation:**

```
K17  =SUMPRODUCT(K4:K14,L4:L14)     (theory: each net times its probability, added
up)
K18  =AVERAGE(E2:E10001)           (simulated average net)
K19  =COUNTIF(E2:E10001,">0")/10000
```

6. **Independence checks** (rows 22–29). Simulated values in column K, exact values in column L. The pattern for a conditional probability is “count both, divide by count of the given”:

```
K23  =COUNTIFS($B$2:$B$10001,6,$C$2:$C$10001,6)/COUNTIF($B$2:$B$10001,6)
```

   Adapt it for each row. For `P(win | doubles)` use

```
K29  =SUMPRODUCT(($B$2:$B$10001=$C$2:$C$10001)*
($E$2:$E$10001>0))/SUMPRODUCT(($B$2:$B$10001=$C$2:$C$10001)*1)
```

7. **Law of Large Numbers chart:** select column G (Running average) → **Insert › Chart** → Line chart. Add a title and axis labels. Re-roll a few times and watch the left end versus the right end.

Desmos alternative (dice only). At desmos.com/calculator, type each line (Desmos formats `random`, `mean`, `<=` as you type). Lists max out at 10,000 entries. The dice icon at the top of the expression list re-rolls everything.

```
d_1 = random([1...6], 10000)
d_2 = random([1...6], 10000)
s = d_1 + d_2
w = {s = 2: 2, s = 12: 2, 5 <= s <= 8: -1, 1}
mean(w)                                  ← simulated average net
histogram(s, 1)                          ← distribution of the sum
mean({d_2 = 6: 1, 0}[d_1 = 6])           ← P(die 2 = 6 | die 1 = 6)
A = [mean(w[1...n]) for n = [100, 200 ... 10000]]
([100, 200 ... 10000], A)                ← running average: the LLN picture
```

Read `[d_1 = 6]` as "keep only the rolls where die 1 was 6." `{condition: 1, 0}` turns each roll into a 1 or a 0, so its mean is a proportion.

Day 2 build — poker and blackjack

Poker tab: 5,000 hands, dealt two ways

The gray engine deals every hand twice. The **correct deal** (columns B–G) draws 5 different cards — *without replacement*, like a real deck. The **bad deal** (columns M–R) uses `RANDBETWEEN(1,52)` for each card — *with replacement*, the shortcut from W3. Look at a few rows of each before you build.

1. **Pay every hand.** `VLOOKUP` finds the hand's name in the paytable (`Y4:Z12`) and returns the net:

   ```
   H2  =VLOOKUP(G2,$Y$4:$Z$12,2,FALSE)        S2  =VLOOKUP(R2,$Y$4:$Z$12,2,FALSE)
   ```

2. **Does card 2 match card 1's rank?** A 1/0 indicator — its average is a proportion:

   ```
   K2  =IF(I2=J2,1,0)        W2  =IF(U2=V2,1,0)
   ```

   Fill H, K, S, W down to row 5001 (copy row 2, Name Box `H3:H5001`, paste — and so on).

3. **Paytable summary** (rows 4–12). Column AA gives the exact number of 5-card hands of each type (out of 2,598,960):

   ```
   AB4  =AA4/2598960     AC4  =COUNTIF($G$2:$G$5001,Y4)     AD4  =AC4/5000
   AE4  =COUNTIF($R$2:$R$5001,Y4)     AF4  =AE4/5000
   ```

   Fill down to row 12; row 13 totals are your check.

4. **Expected value:** `AB15 =SUMPRODUCT(AB4:AB12,Z4:Z12)`, then `AB16 =AVERAGE(H2:H5001)` and `AB17 =AVERAGE(S2:S5001)`.

5. **With vs. without replacement** (rows 20–22). Fill all four columns: simulated correct deal, simulated bad deal, and the exact values — `=3/51` and `=4/52` for row 20; `=1-1317888/2598960` and `=1-(13*12*11*10*9)/13^5` for row 21 (one minus "all five ranks different"); `0` and `=1-(52*51*50*49*48)/52^5` for row 22. For the simulated row 21, add the counts of the hands that contain a repeated rank (one pair, two pair, three of a kind, full house, four of a kind) and divide by 5000.

Blackjack tab: two strategies, same cards

Each row is one hand. The engine plays it twice with the *same* cards: **Strategy A** hits until 17+ (your table's rule) and **Strategy B** hits until 12+ ("never bust"). Comparing on identical cards is a matched-pairs design — chance differences in the cards cancel out. Cards are dealt *with replacement* on purpose: casinos deal from a 6-deck shoe, where one card barely changes the odds (analysis question 7).

1. **Indicator and difference columns:**

   ```
   H2  =IF(G2>21,1,0)          K2  =I2-J2
   ```

   Fill both down to row 5001.

2. **Outcome table** (N8:O12). Strategy A uses column I, Strategy B column J:

   ```
   N8  =COUNTIF($I$2:$I$5001,">0")/5000      N9  =COUNTIF($I$2:$I$5001,0)/5000
   N10 =COUNTIF($I$2:$I$5001,"<0")/5000      N11 =AVERAGE($I$2:$I$5001)
   N12 =COUNTIF($E$2:$E$5001,">21")/5000     (P(player busts) uses the totals in E; B
   uses F)
   ```

   Then `N14 =AVERAGE(K2:K5001)`.

3. **Dealer-bust checks:** `N18 =AVERAGE(H2:H5001)`, then

   ```
   N19  =COUNTIFS($B$2:$B$5001,">=2",$B$2:$B$5001,"
   <=6",$H$2:$H$5001,1)/COUNTIFS($B$2:$B$5001,">=2",$B$2:$B$5001,"<=6")
   ```

   Build N20 for up cards 7–10 or Ace (1). Hint: two COUNTIFS on top, two COUNTIFs on the bottom.

4. **Experiment:** re-roll 5 times and record `N14` each time. Then change the blue cell `N4` to 13, 14, 15, and 16 and record Strategy B's average net for each.

RE-ROLL	1	2	3	4	5	B STANDS ON	13	14	15	16
Avg (A – B)						B avg net				

Dashboard tab

Type your table's rounds and total net and the class totals into the yellow cells. Simulation and theory fill in from your other tabs. For blackjack, the exact theory value for Strategy A comes from a computer program that adds up every possible hand: **E(net) = −0.031 chips per hand**.

Analysis questions

Answer in complete sentences, **in context** (chips, rolls, hands — not just numbers). Cite numbers from your workbook. Due Monday, November 16, with your Sheet shared to Mr. Avendano.

1 Is the dice simulation working?

SIMULATION VS. THEORY

In your Dice distribution table, which sum's simulated relative frequency is farthest from its theoretical probability? Is that gap evidence that the simulation is broken? Explain using the idea of sampling variability. (Re-roll once and see whether the same sum is still the farthest.)

2 A random variable and its expected value

PREVIEW 2.8–2.9

Let X = the Player's net chips on one round of The Field. (a) Write the probability distribution of X as a table with three values. (b) Show that $E(X) = -1/18 \approx -0.056$. (c) Interpret $E(X)$ in context. (d) A classmate says, “So I lose 0.056 chips every roll.” Correct the statement.

3 Four sample sizes

LAW OF LARGE NUMBERS

Describe your running-average chart: what happens on the left (few rolls) compared with the right (thousands)? Then rank your table's average (20 rounds), the class average, and the simulation average by how close you'd *expect* each to be to $E(X)$, and explain. Why doesn't a casino worry when one table walks away ahead?

4 Which dice events are independent?

2.6–2.7

Use exact values (and your simulated values as evidence) to decide whether each pair of events is independent. Write each as a comparison of `P(A | B)` with `P(A)`.

a. “Die 1 is 6” and “die 2 is 6”

b. “Die 1 is 6” and “the sum is 7”

c. “Die 1 is 6” and “the sum is 8”

d. “Doubles” and “the Player wins”

One of these surprises most people. Which one, and why does it work out that way?

5 Inside the poker paytable

EXPECTED VALUE

(a) Report `E(net)` for Five-Card Deal from your summary and interpret it. (b) Multiply each hand's net by its probability. Which *single* hand contributes the most to the Player's expected value — the rare jackpot hands, or something more common? (c) What fraction of hands lose? push? win?

6 Catch the bad deal

WITH VS. WITHOUT REPLACEMENT

Compare the correct deal with the bad deal on four measures: P(card 2 matches card 1's rank), P(at least two cards share a rank), P(the same card appears twice), and average net. Using the word *independent*, explain why the bad deal produces more pairs. If a casino trusted the bad simulation, what decision would it make about this game — and what would happen to its money?

7 When is “with replacement” good enough?

MODELING

A 6-deck blackjack shoe has 312 cards and 24 aces. Find `P(card 2 is an ace | card 1 is an ace)` for the shoe and compare it with `P(ace) = 24/312`. Do the same comparison for a single 52-card deck. Use the two comparisons to explain why the blackjack simulation may deal with replacement but the poker simulation may not.

8 Mimic the dealer, or never bust?

COMPARING WITH SIMULATION

(a) Report the simulated average net for Strategies A and B. Is A's value consistent with the exact `−0.031`? (b) Which strategy is better for the Player, and by about how much per hand? (c) Look at your five re-rolls of `N14`. Are 5,000 hands enough to be *convinced* of the answer to (b)? (d) Which stand-on number from 12–17 looks best?

9 Same rules, unfair game

MULTIPLICATION RULE

Under Strategy A, the Player and the House follow identical rules — yet the House wins in the long run. Find the one situation in the rules that treats the two sides differently. Use your simulated `P(player busts)` and the multiplication rule for independent events to estimate how often that situation happens. Compare your estimate with the size of the House's edge. (What does the +2 for a natural do?)

10 Reading the dealer's up card

CONDITIONAL PROBABILITY

Compare `P(dealer busts)`, `P(dealer busts | up card 2–6)`, and `P(dealer busts | up card 7–A)`. Are “dealer busts” and “up card is 2–6” independent? Our fixed strategy ignores the up card — how could a smarter player use this?

11 Your verdict

COMMUNICATING

Rank the three games (blackjack with Strategy A) from best to worst *for the Player* using expected value. Then: if you could play only **10 rounds**, would you still choose the same game? What about **10,000 rounds**? Justify using both expected value and variability (how spread out the results are — look at the payout tables).

How this lab is scored

Each criterion is scored on the 4-level scale (half levels allowed). Your lab score is the average of the four, to one decimal. **3 = Proficient is the expected standard.**

CRITERION	4 · ADVANCED	3 · PROFICIENT	2 · DEVELOPING	1 · BEGINNING
Data & simulation	Complete paper records; every sim tab works over the full row count and re-rolls; extra exploration (re-roll trials, stand-on sweep) is recorded and used.	Complete paper records; all built columns are formulas filled over the full range; summaries reference the right ranges.	Records or one sim tab incomplete, or formulas cover only part of the range / contain typed-in numbers.	Little usable data; simulation not built or not working.
Probability & expected value	Exact distributions and E(net) correct for dice and poker, shown clearly; uses the paytable contributions to explain *where* the edge comes from.	Exact distributions and E(net) correct and shown, with minor slips at most.	Setup mostly right but with errors in the probabilities or the weighted sum.	E(net) missing, or computed as a plain average of payouts.
Independence & conditional reasoning	Every independence claim is justified by comparing P(A | B) with P(A); explains the dice "sum is 7" surprise and the with/without-replacement difference precisely, including when replacement is a fair model.	Independence decisions correct and justified with conditional probabilities; with/without replacement difference correctly explained.	Some correct decisions, but justification is vague ("they don't affect each other") or a conditional probability is computed incorrectly.	Independence confused with mutually exclusive, or not addressed.
Communicating in context	Interpretations are in context and precise about long-run vs. single rounds; uses variability (re-rolls, sample sizes) to say how convincing each comparison is.	Interpretations are in context; simulated results are compared to theory with appropriate language ("about," "close to," sampling variability).	Numbers reported with little interpretation, or claims like "you lose 0.056 every time."	Answers missing or not connected to the games.

Before you submit: share your Sheet with `wavendano@cushmanschool.org` (Viewer is fine), put the link at the top of your answers, and make sure the Dashboard is filled in. Partners may share one workbook; each student writes their own answers.